

RESEARCH ARTICLE

TEDR-YNet: A Dual-Encoder Retinal Multi-Lesion Segmentation Network

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ABSTRACT Multi-lesion segmentation in diabetic retinopathy faces numerous challenges, including complex fundus structures, small inter-class differences between lesions, blurred lesion boundaries, and class imbalance. To address these challenges, this paper proposes TEDR-YNet, an innovative Y-shaped dual-encoder network for the segmentation of diabetic retinopathy lesions. This network utilizes a dual-encoder architecture to capture features from different representations, significantly enhancing the model's expressive capability. In the bottleneck layer, a meticulously designed lesion-aware enhancement module is introduced to improve multi-scale lesion perception and boundary segmentation accuracy. A skip connection mechanism with gated convolution is innovatively incorporated to strengthen the model's focus on lesion regions. An edge enhancement module is further employed to optimize boundary segmentation performance, and a hybrid loss function is designed to effectively mitigate the class imbalance problem. Extensive experiments conducted on three public datasets IDRID, DDR, and TJDR demonstrate the superior performance of TEDR-YNet, achieving average Dice coefficients of 68.61%, 58.51%, and 58.14%, respectively. These results represent improvements of 11.47%, 7.15%, and 9.8% over the baseline UNet model. Additionally, the mIOU metrics on the three datasets reached 53.24%, 46.84%, and 44.43%, respectively. These findings robustly demonstrate the precision and robustness of TEDR-YNet in multi-lesion segmentation for diabetic retinopathy, offering strong technical support for clinical diagnostic assistance.

INDEX TERMS Lesion segmentation, dual-encoder, gated convolution skip connection, edge enhancement.

I. INTRODUCTION

Diabetic retinopathy (DR) is the leading cause of blindness among working-age populations. In its early stages, the disease typically presents no obvious symptoms, and with the time vision loss and visual field defects occur, it has often progressed to a more severe stage, imposing a heavy burden on patient quality of life and healthcare resources. Therefore, early and accurate screening and diagnosis of diabetic retinopathy are crucial. Color fundus imaging, as a non-invasive and convenient detection method, has become the core technology for large-scale DR screening. However, manual analysis of numerous fundus images by professional ophthalmologists to identify and segment various pathological lesions (such as microaneurysms, hemorrhages, hard exudates, and soft exudates) is a time-consuming and highly

subjective task that cannot meet the growing screening demands.

Diabetic retinopathy is primarily classified into proliferative and non-proliferative lesions based on pathological characteristics. Non-proliferative lesions represent the early stage of the disease, where microaneurysms and hemorrhages appear in the fundus. These lesions are relatively small, and patients do not experience significant discomfort at this stage. As the disease progresses, exudates and cotton wool spots appear in the fundus, leading to vision decline and blurred vision. When new blood vessels develop, the disease advances to proliferative retinopathy, potentially risking blindness.

Traditional segmentation methods primarily rely on thresholding [1], region growing [2], and edge detection [3]. Although these techniques are computationally simple and fast, they are highly sensitive to image quality and susceptible to variations in illumination and noise. Consequently, they

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struggle to handle cases with blurry lesion boundaries or complex grayscale variations, resulting in poor robustness and an inability to meet current segmentation demands. In recent years, intelligent diagnostic systems based on deep learning have demonstrated significant potential in retinal analysis, driving progress in tasks such as optic disc and cup segmentation [4], retinal vessel segmentation [5], and fundus lesion segmentation [6], thereby making automated lesion segmentation a research hotspot. Despite these advancements, the inherent complexity of fundus images, substantial inter-individual variability, and susceptibility to imaging conditions present ongoing challenges. For instance, factors such as non-uniform illumination, occlusion by the optic disc and vessels, and the diversity in lesion morphology, size, and location pose severe obstacles to the precise segmentation of lesions.

CNN-based encoder-decoder architectures are currently among the most mainstream segmentation frameworks. These networks, such as the classic U-Net and its numerous variants, gradually extract multilevel features from images through the encoder part, capturing information from rough contours to fine details, and then progressively upsample and fuse these features through the decoder part to ultimately generate high-resolution segmentation masks. U-Net has been widely favored for its simple structure, relatively small parameter count, and excellent performance in medical image segmentation. Its improved versions (such as those introducing skip connections to better preserve spatial information or adopt deeper encoders to enhance feature representation capabilities) have also been extensively applied in fundus image analysis.

Despite the remarkable progress of existing deep learning methods, the inherent complexity of fundus images continues to pose significant challenges to the precise segmentation of lesions. Conventional single encoder-decoder architectures, in particular, exhibit pronounced limitations when handling Diabetic Retinopathy (DR) lesion segmentation tasks. First, a single encoder typically extracts features directly from the original RGB fundus image. While RGB images contain rich color and texture information, the color contrast between early-stage micro-lesions, such as microaneurysms and small hemorrhages, and the surrounding healthy tissue is often subtle. This makes it difficult for the encoder to consistently capture these critical features, especially under conditions of non-uniform illumination or vascular occlusion. Second, the encoder's progressive downsampling process, designed to extract high-level semantic features, inevitably leads to the loss of crucial spatial detail. This loss is particularly detrimental to segmentation tasks that require accurate lesion boundary delineation. When lesions are irregularly shaped or have ill-defined boundaries, the decoder, relying solely on the features transmitted by the encoder, struggles to achieve precise pixel-level segmentation. Third, a single encoder cannot simultaneously balance the extraction of global contextual information and local fine-grained structures, leading to a performance bottleneck for the model. Notably,

among the three channels of RGB fundus images, the green channel offers a distinct advantage for the feature representation of DR-related lesions. The grayscale contrast between lesions like microaneurysms and hemorrhages and the surrounding tissue is significantly amplified in the green channel. Compared to the red channel, which is susceptible to interference from vessel coloration, and the blue channel, which typically has higher noise content, the green channel effectively suppresses background interference and accentuates the local texture and edge features of lesions. This provides a crucial foundation for optimizing feature extraction strategies.

To overcome these shortcomings, this paper proposes a dual-encoder feature fusion segmentation network, aiming to provide the decoder with two different forms of feature representations, thereby enriching feature representation and enhancing the model's expressive power and robustness. First, to address the difficulty in segmenting tiny lesions, especially microaneurysms, a lesion-aware enhancement module is designed at the network bottleneck layer, significantly improving the model's perception capability for multi-scale lesions and boundary segmentation accuracy. Second, to alleviate the problem of small inter-class differences between lesions, an innovative skip connection with gated convolution is proposed, leveraging the concept of LBP [7] to enhance the model's focus on lesion regions, capture micro-texture differences in lesion areas, and suppress the generation of non-lesion pixels. Next, to enhance the segmentation effect of lesion edges, an edge enhancement module is proposed, which enhances lesion edges after each upsampling layer by simulating the visual system's sensitivity to local contrast. Finally, to address the class imbalance problem, a hybrid loss function is designed by combining the advantages of Dice loss and categorical cross-entropy loss, enhancing the model's segmentation capability for minority lesion samples. The contributions of this paper are:

- A dual-encoder end-to-end multi-class segmentation network, TEDR-YNet, is proposed, effectively improving the segmentation performance for four different types of lesions, especially microaneurysms.
- A module convolution based on LBP principles is designed and integrated into skip connections, which enhances important features while suppressing the generation of irrelevant features.
- An innovative edge enhancement module is developed to enhance the semantic information of lesion edges, adaptively enhancing the structural details of lesions by simulating the visual system's sensitivity to local contrast.
- Excellent segmentation results have been demonstrated on three diabetic retinopathy datasets: IDR1D, DDR, and TJDR.

The remainder of this paper is organized as follows. Section II reviews existing lesion segmentation methods and dual-encoder architectures. In Section III, the proposed

TEDR-YNet model architecture and preprocessing methods are elaborated in detail. Section IV introduces the datasets and evaluation metrics. In Section V, the TRDR-YNet model is compared with other state-of-the-art methods. In Section VI, ablation studies are conducted to collectively verify the effectiveness and robustness of the model's key components, as well as their ability to improve computational efficiency while maintaining performance. Finally, Section VII provides the conclusion.

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II. RELATED WORK

A. DEEP LEARNING-BASED SEGMENTATION METHODS

The introduction of Fully Convolutional Networks (FCN) [8] replaced inefficient sliding window approaches by enabling end-to-end pixel-level prediction. Subsequently, U-Net [9], with its symmetric encoder-decoder architecture and skip connections, has become an essential framework in medical image segmentation, particularly crucial for accurately segmenting small pathological targets. The success of U-Net and FCN has spurred numerous improved architectures, significantly enhancing both the accuracy and efficiency of retinal lesion segmentation. In 2019, Xiao et al. [10] proposed a pixel-level end-to-end segmentation system that incorporated adversarial loss into the segmentation loss, improving the lesion segmentation performance of baseline models. In 2022, Huang et al. [11] introduced a deep learning network based on Relation Transformation Blocks (RTB) and Global Transformer Blocks (GTB) for automatically segmenting four types of lesions in diabetic retinopathy (DR). In 2023, Ullah et al. [12] presented a semi-supervised multi-task decoder network that effectively improved segmentation performance by leveraging large amounts of unlabeled data combined with multi-task and semi-supervised learning; however, the use of multiple decoders resulted in high computational complexity and extended training time. In 2024, Xu et al. [13] proposed a novel deep learning approach: the Multi-scale Spatial Attention Gate (MSAG) mechanism network. This model processes input images at different scales to extract a range of semantic information, with the innovative spatial attention gates fusing low-level spatial details with high-level semantic content by assigning hierarchical attention weights to achieve accurate segmentation, though challenges remain in accurately distinguishing between lesions. In 2025, Ma et al. [14] integrated CNN and Transformer in a dual attention module that eliminated semantic inconsistency between the encoder and decoder, better integrating contextual information, but still faced significant challenges in accurately processing smaller lesions. To address the aforementioned issues, this paper proposes a cross-pooling fusion module that enhances the semantic information of lesions while fusing features.

B. DUAL-ENCODER ARCHITECTURE

The development of dual-encoder architecture originated from addressing the limited representation capability of single feature extraction networks in medical image analysis. Traditional single-encoder models like U-Net often struggle to simultaneously capture global structural information and local detailed features when processing complex and variable DR lesions, leading to missed detection of small lesions and misjudgment of ambiguous boundaries. To overcome this challenge, researchers began exploring dual-branch encoder designs to obtain more complementary image representations through parallel feature extraction paths. Early explorations in the field of diabetic retinopathy analysis have primarily focused on innovations in model architecture. For instance, Jian et al. [15] constructed a novel parallel encoder by combining multiple convolutional modules to enrich feature diversity. However, its complex model design limited its generalization capability across diverse datasets. Yin et al. [16] integrated the advantages of U-Net and U2-Net, employing a dual-decoder structure to generate segmentation and boundary maps separately before fusing them. This approach significantly enhanced boundary accuracy but introduced additional computational burden. To balance the extraction of global and local information, Guo and Peng [17] proposed CARNet, which utilizes ResNets of varying depths as dual branches and incorporates an attention mechanism to dynamically fuse features, effectively focusing on lesion areas. Nevertheless, its sensitivity to small lesions remains to be improved. Recent research has begun to explore more diverse feature inputs and fusion strategies. In 2023, Pavani et al. [18] introduced RILBP-YNet, which adopts a dual-encoder, single-decoder architecture. One branch processes the original image, while the other extracts texture features from a handcrafted Local Binary Pattern (LBP), effectively enhancing lesion segmentation performance. However, as a handcrafted descriptor, LBP's representational power is limited by its predefined patterns, making it difficult to capture the more complex, high-order semantic information within DR lesions. In 2024, Agarwal et al. [19] combined a deep quasi-recurrent self-attention mechanism with a dual-encoder, which improved segmentation accuracy but also significantly increased the model's computational cost and parameter count. In the same year, Xu et al. [20] proposed a dual-branch, prior-guided attention fusion Transformer model for multi-lesion segmentation in DR. This model fuses lesion and vascular branches through an attention-fusion Transformer encoder, effectively boosting segmentation performance. However, its performance is susceptible to the quality of vascular information extracted by the pretrained model. Building upon the aforementioned studies, this paper proposes a novel dual-encoder feature fusion network. It directly utilizes the original image and the green channel image, which contains abundant pathological information, as inputs for the dual encoders. This design enables the model to autonomously

learn complementary features, thereby enhancing its generalization performance. Concurrently, inspired by the LBP paradigm, we designed a gated convolution-based skip connection to capture local structural patterns in a learnable manner, overcoming the expressive bottleneck of traditional LBP. In summary, this work aims to mitigate the impact of inter-class similarity among lesions on segmentation outcomes, enhance the segmentation performance for small lesions, and simultaneously reduce computational costs.

III. SEGMENTATION NETWORK

This paper proposes a dual-encoder feature fusion network for multi-lesion segmentation of diabetic retinopathy. The network primarily consists of a dual-branch encoder (DE), Progressive Attention Fusion Module (PAF), lesion perception enhancement module (LPE), Edge Enhancement Module (LEM), and skip connections with gated convolution (LSG). The architecture is illustrated in Figure 1, with detailed descriptions of each module as follows.

A. DUAL ENCODER BRANCHES

Color fundus images contain rich color information across red, green, and blue channels, which can reveal multi-level features of lesions such as blood vessels, hemorrhages, and microaneurysms. This information is crucial for the accurate diagnosis of diabetic retinopathy. Hemoglobin exhibits the highest absorption rate in the green channel, which results in significant contrast for hemorrhages and microaneurysms. To extract complementary features from both color fundus images and green channel images, this paper proposes a dual-encoder architecture. For the color fundus images, we designed improved residual blocks with downsampling, as illustrated in Figure 2. Meanwhile, the green channel encoder utilizes the residual modules from EfficientNetV2-S [21] for downsampling. As an efficient convolutional neural network architecture, the residual modules of EfficientNetV2-S are meticulously designed to significantly reduce computational complexity while maintaining high accuracy.

The input features $x \in \mathbb{R}^{C \times H \times W}$ first undergo channel compression to one-fourth of their original dimension through 1×1 convolution, followed by batch normalization and GELU activation function processing, which significantly reduces subsequent computational complexity. Subsequently, the features pass through a depth-wise separable convolution with a kernel size of 7×7 , efficiently capturing long-range relationships and complex spatial patterns between pixels. The channels are then expanded to the target dimension, integrating information across all channels. Finally, the processed features are added element-wise to the input, preventing the loss of underlying information during transmission between different network layers and effectively mitigating the vanishing gradient problem.

B. PROGRESSIVE ATTENTION FUSION MODULE

Lesions in diabetic retinopathy exhibit high diversity and complexity, including microaneurysms, hemorrhages, hard

exudates, and soft exudates. These lesions demonstrate significant differences in color, morphology, texture, and contrast with the background. Conventional feature fusion methods simply concatenate and fuse features from different sources such as RGB and green channels, failing to adaptively perform differential processing for the unique visual characteristics of each lesion type. For instance, hemorrhages and microaneurysms show higher contrast in the green channel, while the color information in RGB channels is more critical for exudates. To address this limitation, this article proposes an enhanced feature fusion module for DR lesions that operates at the bottleneck layer of the network. The core concept is to assign dedicated feature source weights to each lesion category, constructing specific feature representations, and enhancing the discriminative power of fused features through an attention mechanism, as illustrated in Figure 3.

The features $f_{RGB} \in \mathbb{R}^{C \times H \times W}$ extracted from the RGB encoder and features $f_{Green} \in \mathbb{R}^{C \times H \times W}$ extracted from the green encoder are first concatenated along the channel dimension. Subsequently, a convolution with a kernel size of 3 is applied to enhance the expressive capability of the features. The resulting feature maps are then flattened into vectors with dimensions $B \times N$ for subsequent attention computation. Weights α are computed for the features output from the fully connected layer, and these attention weights α are multiplied with the previously processed multi-source features. The weighted features from all sources are then summed to obtain the weighted fused feature representation. Finally, a 1×1 convolution is employed to integrate the features.

C. LESION PERCEPTION ENHANCEMENT MODULE

In diabetic retinopathy lesion segmentation, different lesions exhibit distinct scale characteristics: microaneurysms are tiny and require high-resolution details; hemorrhage regions have variable sizes and need medium receptive fields; while exudates may form large clusters that require large receptive fields to understand global context. Traditional single-scale convolution kernels struggle to simultaneously capture these multi-scale features. Furthermore, the boundaries between lesions and background are often indistinct, particularly in hemorrhage and soft exudate regions, where simple feature extraction can easily lead to inaccurate boundary predictions. To address these challenges, this paper proposes a Lesion Perception Enhancement Module, which significantly improves the model's perception capability for multi-scale lesions and boundary segmentation accuracy through a triple mechanism of parallel multi-scale feature extraction, explicit edge enhancement, and adaptive feature calibration. As illustrated in Figure 4, Initially, the module concurrently extracts features at varying receptive fields through parallel 1×1 , 3×3 (with a dilation rate of 6), and 5×5 (with a dilation rate of 12) convolutions. This design ensures the capture of fine-grained, local medium-scale, and large-scale structural information. Subsequently, channel-wise global context, generated via global average

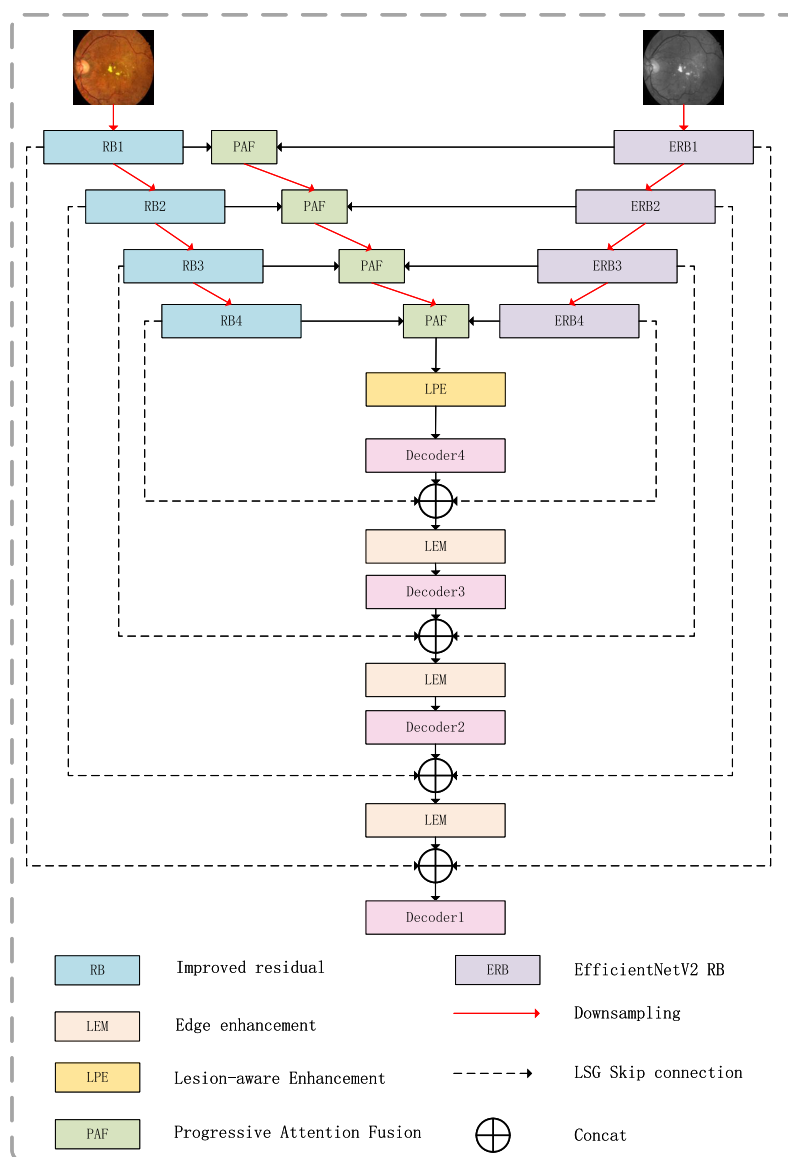


FIGURE 1. Dual-branch encoder-based multi-class lesion segmentation network.

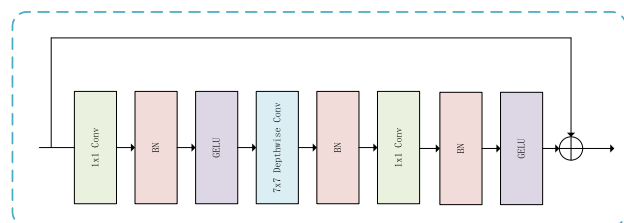


FIGURE 2. RB module.

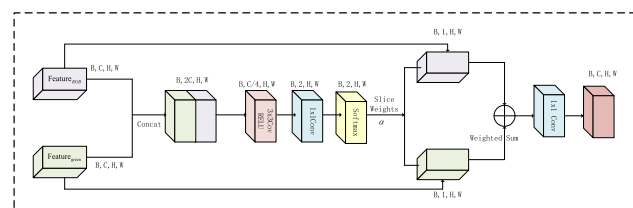


FIGURE 3. PAF module.

pooling, is processed through a 1×1 convolution and a Sigmoid function to produce a global attention map. This map then re-weights the aforementioned multi-scale features on a per-pixel basis, facilitating a synergistic enhancement

of global and local information. The re-weighted features are concatenated along the channel dimension and then passed through a 1×1 convolution to compress the channel count, thereby unifying the multi-scale features. To further

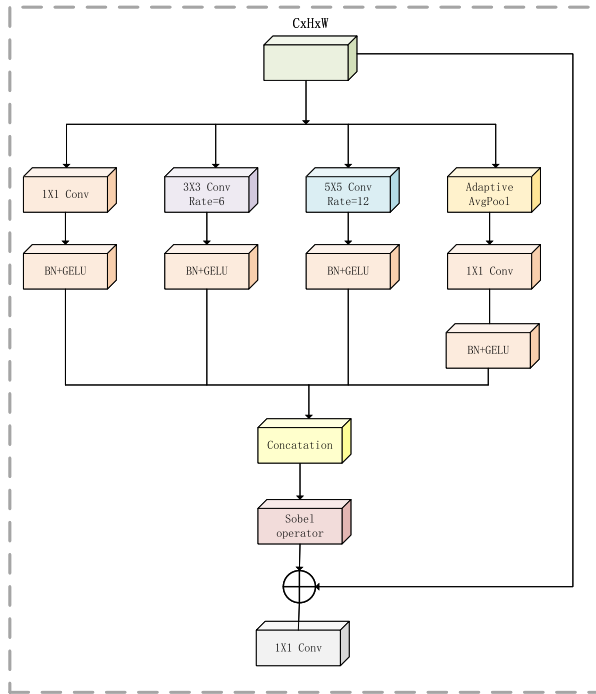


FIGURE 4. LPE module.

accentuate lesion boundaries, the fused features undergo a learnable Sobel edge filtering, implemented via depth-wise separable convolution. The resulting edge attention map is multiplied with the features and added via a residual connection, which significantly improves the discriminability of fine details such as micro-hemorrhages and soft exudates. Finally, the module concatenates the original input features with the enhanced features along the channel dimension, and a subsequent 1×1 convolution restores the channel count to produce the residual-fused output.

D. SKIP CONNECTIONS WITH SEMANTIC GATING

To enhance the model's focus on lesion regions, suppress the generation of non-lesion pixels, and improve the discriminative ability to distinguish between different lesions, a lightweight semantic gating module called LBP-Semantic-Gate (LSG) was designed, inspired by the core concept of Local Binary Pattern. This module is embedded in the skip connections between the encoder and decoder, with its output serving as a spatial attention gating map that is element-wise multiplied with the corresponding upsampled feature maps from the decoder, thereby adaptively calibrating and enhancing the features.

Traditional skip connections directly transfer multi-scale features from the encoder to the decoder without adequately emphasizing the differences between lesion regions and normal tissues. In the high-level features extracted by the encoder, each channel carries different semantic information. A "lesion region" can be defined at the feature level as an area where its local contextual features (neighborhood) exhibit significant differences from the global contextual

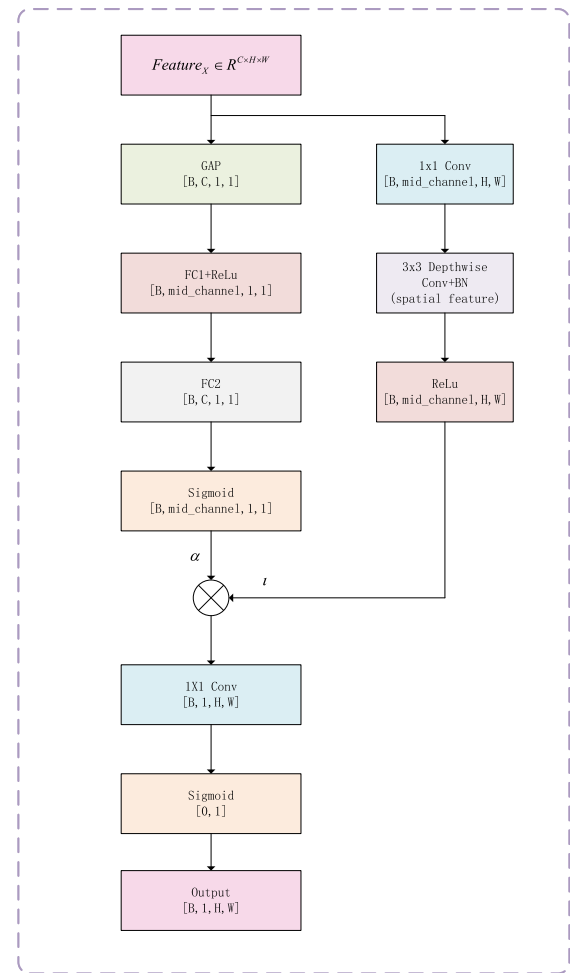


FIGURE 5. LSG module.

semantics (center). The core task of the LSG module is to quantify this difference and generate a weight map accordingly, which amplifies lesion features while suppressing non-lesion features during the decoding process. The model diagram is shown in Figure 5.

The input feature $x \in \mathbb{R}^{C \times H \times W}$ is processed in parallel through global and local pathways. The global pathway extracts channel-level global context information through Global Average Pooling (GAP) and a two-layer fully connected network, generating channel attention weights α . The local pathway extracts local spatial features through 1×1 convolution for dimensionality reduction and depth-wise separable convolution, obtaining local feature representations l . The global attention α is projected into the same channel space as the local features, and the projected global attention is fused with the local features through element-wise multiplication, simulating the comparison operation of LBP. A 1×1 convolution maps the fused features to a single-channel space, and a Sigmoid function is applied to generate a spatial gating map with values in the range $[0, 1]$. This gating map is then multiplied with the decoder's hierarchy to enhance lesion region features while suppressing background regions.

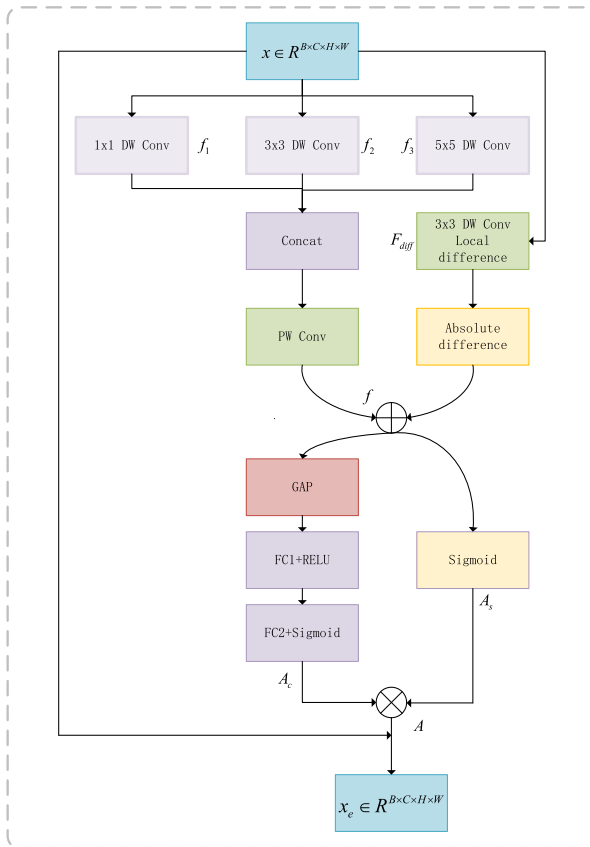


FIGURE 6. LEM module.

E. EDGE STRUCTURE FEATURE ENHANCEMENT MODULE

To address the inherent challenges of diabetic retinopathy lesions, including their small morphology, blurred edges, and low contrast with the background, this paper proposes a novel Edge Structure Feature Enhancement Module (LEM) designed to explicitly strengthen the network's perception and representation capabilities for lesion structure and edge information. The core concept of this module is to adaptively enhance diagnostically significant structural details in a differentiable manner by simulating the visual system's sensitivity to local contrast. The model diagram is illustrated in Figure 6.

First, the input feature x is simultaneously processed using three convolutional kernels of different sizes (f_1, f_2, f_3), followed by channel concatenation of the resulting multi-scale features. A pointwise convolution (PW Conv) is then utilized for fusion and dimensionality reduction, generating a feature representation rich in multi-scale contextual information. Subsequently, a local difference operator is introduced to explicitly enhance the gradient information between lesion edges and the background. This operator calculates the feature differences between the central pixel and surrounding pixels through a 3×3 depth-wise separable convolution (DW Conv), thereby highlighting local region contrast. The multi-scale contextual features and local difference features

are then added element-wise, yielding feature f which contains both rich contextual information and emphasized local structural variations. To further enhance the model's representation capability, a lightweight channel attention sub-module is introduced to generate a channel attention map A_c . Meanwhile, the fused features pass through a Sigmoid activation function to produce a spatial attention map A_s . These two attention maps are then fused into A . Finally, through residual learning, the learned attention weights are applied to the original input to generate the enhanced output feature.

F. LOSS FUNCTION

To effectively address the class imbalance problem in multi-class lesion segmentation of diabetic retinopathy, this paper proposes a hybrid loss function that combines classification cross-entropy loss with Dice loss. This loss function optimizes both pixel-level classification accuracy and region overlap through weighted fusion of these two complementary loss metrics.

The classification cross-entropy loss [22] serves as the fundamental classification loss, measuring the discrepancy between the predicted probability distribution and the ground truth labels. To address potential class imbalance issues in the dataset, a class weighting mechanism is also introduced.

$$l_{ce} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C \omega_c \cdot y_{i,c} \cdot \log(\hat{y}_{i,c}) \quad (1)$$

Here, N represents the total number of pixels, c denotes the number of classes, ω_c is the weight coefficient for class c , $y_{i,c}$ indicates the ground truth label of pixel i belonging to class c , and $\hat{y}_{i,c}$ represents the corresponding predicted probability.

The Dice Loss [23], based on the Dice coefficient, directly optimizes the overlap between predicted regions and ground truth regions, offering inherent advantages for small object segmentation. For the multi-class segmentation task in this study, we employ a strategy of calculating and averaging the Dice loss for each class individually.

$$l_{Dice} = 1 - \frac{1}{C} \sum_{c=1}^C \frac{2 \sum_i y_{i,c} \cdot \hat{y}_{i,c} + \psi}{\sum_i y_{i,c} + \sum_i \hat{y}_{i,c} + \psi} \quad (2)$$

Here, ψ is a smoothing factor to prevent the denominator from being zero. The combined loss function is constructed by weighted fusion of the aforementioned loss functions.

$$l_{combined} = \alpha \cdot l_{Dice} + \lambda \cdot l_{ce} \quad (3)$$

Here, α, λ are the weights of each loss function, $\alpha = 0.7, \lambda = 0.3$.

IV. EXPERIMENTS

A. DATASET

The IDRID dataset [24] is a specialized dataset for research on diabetic retinopathy (DR) and diabetic macular edema (DME). It consists of 81 fundus images from Indian patients

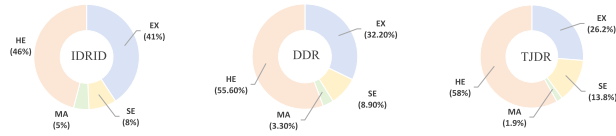


FIGURE 7. (a) Annotated pixel proportion for four lesions in the IDRiD dataset. (b) Annotated pixel proportion for four lesions in the DDR dataset. (c) Annotated pixel proportion for four lesions in the TJDR dataset.

with a resolution of 2848×4288 , each accompanied by expert manual annotations for four types of diabetic retinopathy lesions. The dataset includes 80 images with hard exudates (HE), 40 images with soft exudates (SE), 81 images with microaneurysms (MA), and 81 images with exudates (EX). In the IDRiD dataset, 54 images are used for training and 27 images for testing.

The DDR dataset [25] contains 13,673 color fundus images with resolutions ranging from 1088×1920 to 3456×5184 . Among these, 757 images provide pixel-level ground truth annotations for EX, HE, MA, and SE lesions. In the DDR dataset, 608 images are allocated for training and 149 images for testing.

The TJDR dataset [26] comprises 561 high-quality color fundus images from Tongji Hospital affiliated with Tongji University, including microaneurysms (MA), hemorrhages (HE), hard exudates (EX), and soft exudates (SE). Each DR image in this dataset has pixel-level manual annotations by ophthalmologists. In the TJDR dataset, 448 images are used for training and 113 images for testing, as shown in Figure 7.

B. EXPERIMENTAL SETUP

The TEDR-YNet model was implemented using the PyTorch framework. Training and evaluation were conducted on a Windows platform equipped with a NVIDIA RTX 4060Ti GPU (32 GB of VRAM). The detailed training hyperparameters are summarized in Table 1.

TABLE 1. Training hyperparameter configuration.

Hyperparameter	Setting
Optimizer	Adam
Initial Learning Rate	1×10^{-4}
Weight Decay	1×10^{-5}
Batch Size	4
Epochs	100
Model Selection	Epoch with minimum validation loss

Prior to training, all fundus images underwent a standardized preprocessing pipeline. First, images from all datasets were resized to 512×512 pixels to unify the input dimensions. The green channel was then isolated for its enhanced contrast in retinal imagery. To make fine pathological details more prominent, we applied Contrast-Limited Adaptive Histogram Equalization (CLAHE) to these green channel images. The resulting enhanced images served as the final input to the network.

C. EVALUATION METRICS

To objectively evaluate the proposed method, we compare the predicted segmentation results with their corresponding ground truth labels by calculating the Area Under the Precision-Recall Curve (AUPR), Dice Coefficient, and Intersection-Over-Union (IOU). (1) AUPR: Precision and recall are plotted as variables on a two-dimensional plane. The area under the curve represents the AUPR value, where a higher AUPR indicates better model performance. (2) Dice: Measures the balance between precision and recall in segmentation models. (3) IOU: Evaluates the degree of overlap between predicted segmentation maps and ground truth labels. It is defined as follows.

$$Dice = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (4)$$

$$IOU = \frac{TP}{TP + FP + FN} \quad (5)$$

Here, true positives (TP) and true negatives (TN) represent the number of correctly segmented lesion pixels and non-lesion pixels, respectively; false positives (FP) and false negatives (FN) represent the number of incorrectly segmented lesion pixels and non-lesion pixels, respectively.

V. RESULTS

A. QUANTITATIVE ANALYSIS WITH STATE-OF-THE-ART METHODS

Lesion segmentation experiments were conducted on popular networks, including U-Net [9], UNet++ [27], and HR-Net [28]. To eliminate the influence of preprocessing procedures on experimental results, all models employed identical preprocessing methods, with experimental details configured to be as consistent as possible with this study. Furthermore, we compared our approach with currently advanced segmentation methods on the IDRiD and DDR datasets, including EAnet [29], SegDRoWS [30], RILBP-YNet [18], HRDecoder [31], HACDR-Net [32], L-seg [33], and SEBNet [34]. Additionally, on the TJDR dataset, we performed a comparison with the latest model, α -Fed [35].

Tables 2, 3, and 4 present the qualitative segmentation results of four lesions across three diabetic retinopathy datasets: IDRiD, DDR, and TJDR. As shown in the tables, TEDR-YNet achieves the best segmentation performance for microaneurysms (MA) across all three datasets, with AUPR scores of 56.37%, 50.78%, and 50.61% respectively, representing improvements of 7.11%, 27.3%, and 2.41% compared to the second-best methods SEBNet, EAnet, and HRNet.

On the IDRiD dataset, the Dice coefficients for exudates (EX) and soft exudates (SE) (81.91% and 75.13%) outperform the second-best method HRDecoder by 1.3% and 2.14% respectively. Additionally, the IOU values for MA, EX, and SE (36.97%, 69.37%, and 60.17%) surpass the second-best method HACDR-Net by 4.42%, 1.83%, and 2.54% respectively.

TABLE 2. Comparison of TEDR-YNet with state-of-the-art methods on the IDRID dataset (Bold indicates the best performance).

Model	AUPR				DICE				IOU				Param (M)
	MA	HE	EX	SE	MA	HE	EX	SE	MA	HE	EX	SE	
UNet	44.26	53.71	77.24	67.19	46.17	48.06	68.55	65.77	30.42	34.22	53.52	39.65	13.40
UNet++	37.88	55.13	79.56	44.75	47.23	52.92	77.62	58.47	31.32	38.37	52.57	42.71	36.36
HRNet	45.67	57.62	77.32	49.89	48.08	57.88	62.83	63.23	34.50	41.05	46.64	57.00	9.64
EANet	48.93	49.97	69.72	43.44	34.75	56.42	72.71	49.92	21.12	39.54	58.21	35.84	34.20
RILBP-YNet	38.04	69.14	78.58	72.67	36.02	66.54	76.32	49.88	21.97	49.85	61.70	33.22	4.80
HRDecoder	49.16	70.80	87.55	77.65	49.35	66.68	80.61	72.99	32.55	50.04	67.54	57.63	9.64
HACDR-Net	43.02	68.50	86.90	76.73	44.12	64.27	78.50	71.96	36.17	47.35	64.61	56.21	—
SegDRoWS	—	—	—	—	48.95	68.01	79.42	70.73	32.40	51.52	65.86	54.71	2.27
SEBNet	49.26	69.71	88.36	76.69	35.92	61.87	79.84	71.87	21.89	44.80	66.45	56.11	15.70
Our Method	56.37	65.26	84.30	77.28	53.98	63.43	81.91	75.13	36.97	46.45	69.37	60.17	16.71

TABLE 3. Comparison of TEDR-YNet with state-of-the-art methods on the DDR dataset (Bold indicates the best performance).

Model	AUPR				DICE				IOU				Param (M)
	MA	HE	EX	SE	MA	HE	EX	SE	MA	HE	EX	SE	
UNet	29.78	44.09	58.36	60.97	39.44	55.51	69.63	40.87	32.61	45.45	62.14	28.16	13.40
UNet++	30.33	42.99	60.07	58.61	46.67	53.26	68.84	43.06	39.90	43.25	61.72	30.43	36.36
HRNet	31.75	43.66	58.87	62.85	41.07	53.54	49.03	33.21	28.67	38.94	37.94	20.53	9.64
L-seg	10.52	35.86	55.64	26.48	23.01	41.53	54.44	28.01	10.13	24.23	35.24	28.01	14.30
EANet	32.41	37.01	48.68	66.00	40.31	42.79	67.98	41.41	25.39	27.58	52.56	28.54	34.20
HRDecoder	24.60	55.40	64.84	51.93	27.92	52.70	60.67	51.57	16.22	35.03	43.54	34.22	9.64
HACDR-Net	24.26	55.02	65.15	56.75	26.34	53.96	61.24	55.64	15.17	36.95	44.13	38.54	—
SegDRoWS	—	—	—	—	11.78	33.20	43.10	33.50	21.07	49.85	60.24	50.18	2.27
SEBNet	15.78	46.31	65.30	43.72	16.03	44.40	52.69	44.36	8.72	28.53	35.77	28.50	15.70
Our Method	50.78	55.56	67.85	67.09	47.54	65.55	65.85	55.10	35.55	51.14	63.27	37.38	17.20

TABLE 4. Comparison of TEDR-YNet with state-of-the-art methods on the TJDR dataset (Bold indicates the best performance).

Model	AUPR				DICE				IOU				Param (M)
	MA	HE	EX	SE	MA	HE	EX	SE	MA	HE	EX	SE	
UNet	31.65	50.94	68.55	50.57	49.12	40.72	57.29	46.21	24.93	49.51	54.03	43.38	13.40
UNet++	34.33	52.88	69.67	52.03	41.25	40.94	53.93	40.94	28.74	49.49	50.47	44.34	36.36
HRNet	48.20	59.90	70.23	59.28	36.21	38.84	56.33	47.96	31.02	47.51	42.46	39.78	34.20
α -Fed	—	—	—	—	—	—	—	—	21.48	46.60	39.94	38.45	—
Our Method	50.61	60.68	72.60	54.33	49.67	57.30	71.93	53.64	32.23	47.79	57.37	40.33	17.20

On the DDR dataset, TEDR-YNet achieves optimal AUPR values for all four lesion types: MA (50.78%), hemorrhages (HE) (55.56%), EX (67.85%), and SE (67.09%). Notably, the AUPR for MA and HE shows improvements of 18.37% and 18.55% respectively compared to the second-best model EANet (32.41% and 37.01%). Furthermore, the Dice coefficients for MA, HE, and EX (47.54%, 65.55%, and 65.85%) all achieve optimal results. The IOU values for MA, HE, and SE (35.55%, 51.14%, and 37.38%) also reach optimal performance, exceeding the second-best methods UNet, SegDRoWS, and SEBNet by 2.94%, 1.29%, and 8.88% respectively. The IOU for EX is 63.27%, which is only 1.55% lower than the second-best model UNet.

On the TJDR dataset, TEDR-YNet achieves optimal AUPR values for all four lesions: MA (50.61%), HE (60.68%), EX (72.60%), and SE (54.33%), with corresponding Dice coefficients of 49.67%, 57.30%, 71.93%, and 53.64%. Compared to the baseline model UNet, the AUPR improvements are 18.96%, 9.74%, 4.05%, and 3.76% respectively, while the Dice coefficient improvements are 0.55%, 16.58%, 14.64%, and 7.43% respectively. Additionally, the IOU values for MA and SE (32.23% and 40.33%) demonstrate superior

performance, with MA's IOU exceeding the second-best model HRNet by 1.21%, while SE's IOU is 4.01% lower than the second-best model UNet.

The experimental results indicate that TEDR-YNet exhibits excellent performance in the multi-lesion segmentation task for diabetic retinopathy. The model achieves optimal results across multiple evaluation metrics on the IDRID, DDR, and TJDR datasets, demonstrating particularly significant advantages in MA segmentation compared to other methods, and showing strong generalization capabilities and robustness. It is noteworthy that the superior performance of TEDR-YNet is not achieved at the expense of increased model complexity. With only 16.71M parameters, it is substantially smaller than EANet (34.2M) and UNet++ (36.36M), which deliver comparable performance. This result effectively demonstrates the model's efficacy in achieving higher segmentation accuracy while maintaining a lightweight architecture.

B. QUALITATIVE ANALYSIS

To evaluate the performance of the TEDR-YNet network in segmenting retinal lesions, we visualized the segmentation

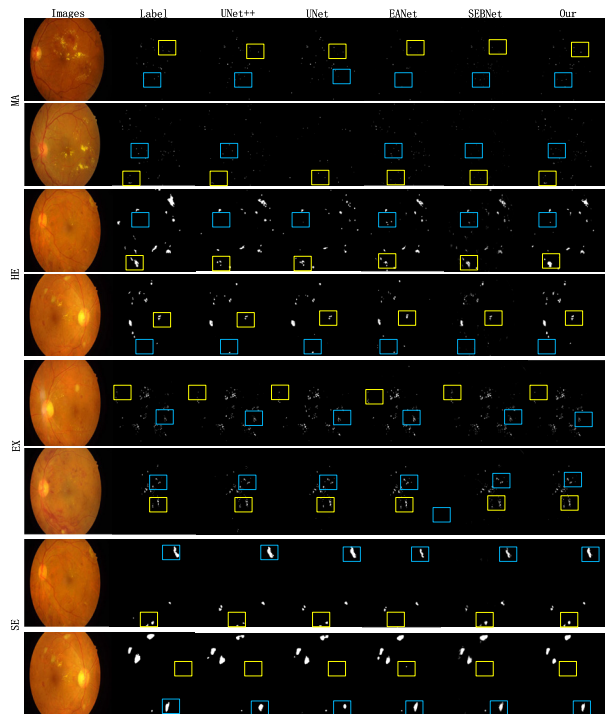


FIGURE 8. Segmentation results of TEDR-YNet on the IDRID dataset.

results of EANet, SEBNet, and TEDR-YNet on the IDRID, DDR, and TJDR datasets to verify the model's effectiveness in segmenting challenging small lesions in retinal images. Fundus images contain numerous small lesions. Therefore, we used colored boxes to emphasize details, where blue boxes indicate areas with incorrect segmentation and yellow boxes indicate regions with missed segmentation.

The segmentation results on the IDRID dataset are shown in Figure 8. By comparing with UNet, UNet++, EANet, and SEBNet, it can be observed that EANet demonstrates excellent performance in segmenting EX but performs poorly in segmenting the other three types of lesions. SEBNet can segment small lesion pixels that are difficult to segment among the four lesion types, but the segmented lesion edges are blurry with numerous missed lesions. To address these issues, our method enhances the segmentation effect of lesion edges through an edge enhancement module and employs a dual-encoder branch to obtain features in two different representation forms. From the segmentation results of the four lesion types, the improvement is significant, especially for MA.

The segmentation results on the DDR dataset are shown in Figure 9. By comparing with UNet, UNet++, and HRNet, it can be seen that our method demonstrates clear advantages in the segmentation of all four lesion types. UNet and UNet++ exhibit obvious missed detections in small lesions, especially microaneurysms, while HRNet, although capable of better capturing lesion regions, preserves insufficient edge details. Our method not only effectively identifies small lesions but also significantly improves the clarity of lesion

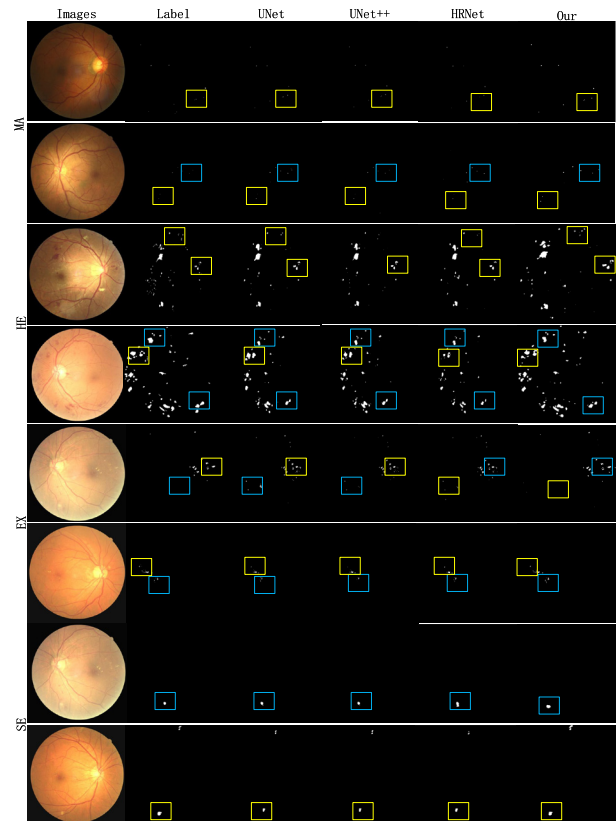


FIGURE 9. Segmentation results of TEDR-YNet on the DDR dataset.

edges, particularly in the segmentation of MA and HE, reducing missed segmentation phenomena and resulting in more complete and accurate segmentation results.

The segmentation results of the TJDR dataset are shown in Figure 10. To more intuitively observe the segmentation effects of the four lesion types, four colors were used to represent different lesion types: yellow for microaneurysms, green for hemorrhages, red for hard exudates, and blue for soft exudates. It can be seen that our proposed method can alleviate interference caused by inter-class similarity and accurately segment lesions of various shapes and sizes from complex fundus images.

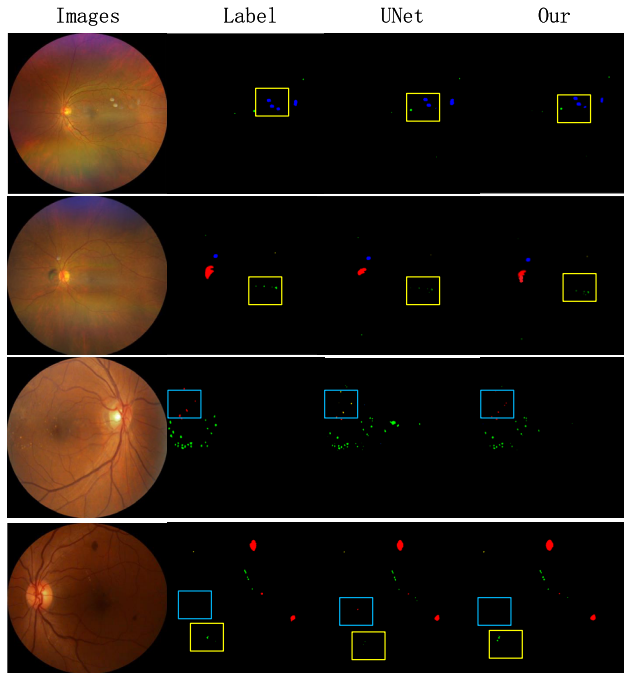
VI. ABLATION STUDY

A. ABLATION STUDY OF NETWORK MODULES

As illustrated in Figure 1, TEDR-YNet is a four-layer dual encoder-decoder Y-shaped architecture composed of a dual-branch encoder, Progressive Attention Fusion Module, lesion perception enhancement module, edge enhancement module, and skip connections with gated convolution. To validate the effectiveness of each component, we conducted ablation experiments using the IDRID dataset to evaluate the impact of individual components on the model. The baseline model for this study is a dual-branch residual network with a VGG16 backbone, where features extracted by the two encoder branches are fused via concatenation. Building upon this

TABLE 5. Results of the ablation experiments on the IDRID dataset (Bold indicates the best performance).

Methods	AUPR				DICE				IOU			
	MA	HE	EX	SE	MA	HE	EX	SE	MA	HE	EX	SE
backbone	45.93	57.85	72.32	68.10	45.24	52.74	71.52	57.38	30.36	40.16	56.33	43.55
+PAF	47.23	58.42	73.02	64.30	46.69	54.35	72.70	62.98	30.57	37.62	58.12	49.76
+LPE	48.89	60.85	75.52	68.80	48.52	57.44	74.54	61.58	32.16	39.82	60.14	53.15
+LEM	54.74	63.44	78.94	76.22	53.53	61.02	81.25	75.97	37.31	42.47	68.42	59.22
+LSG	56.37	65.26	84.30	77.28	53.98	63.43	81.91	75.13	36.97	46.45	69.37	60.17

**FIGURE 10.** Segmentation results of TEDR-YNet on the TJDR dataset.

foundation, we followed a progressive integration strategy, as detailed in Table 5, to sequentially incorporate the PAF, LPE, LEM, and LSG modules. It is crucial to note that during the integration of the PAF module, the encoder for the green channel was concurrently replaced with the EfficientNetV2-S architecture to achieve superior feature downsampling efficiency. To ensure a fair comparison, all experiments were conducted under a uniform hyperparameter configuration. The results clearly demonstrate that the introduction of each module yielded a significant and consistent improvement in model performance.

B. IMPACT OF DIFFERENT INPUTS AND BACKBONE NETWORKS ON SEGMENTATION RESULTS

The nature of input data determines the richness and quality of information that the model can perceive, while the architecture of the backbone network directly influences the model's ability to extract and comprehend high-level semantic features. To systematically evaluate the effects of different input forms and backbone networks on segmentation performance, this study utilized the original images and their green channel images as input sources, and employed ResNet50 [36], VGG16 [37], Vision Transformer

[38], and DeepLabV3+ [39] as green encoder architectures, respectively. Comparative experiments were conducted on the IDRID dataset, with the results presented in Table 6.

C. COMPUTATIONAL EFFICIENCY ANALYSIS

To quantitatively evaluate the computational efficiency of TEDR-YNet and conduct a comprehensive comparison with mainstream baseline models, this section reports the number of floating-point operations (FLOPs) and the inference speed (FPS) for each model. All models were tested under identical conditions on a unified hardware platform equipped with an NVIDIA RTX 4060Ti GPU to ensure fairness.

Table 7 presents the detailed computational efficiency of the models on the IDRID dataset. As shown, the proposed TEDR-YNet demonstrates a balanced and outstanding performance in terms of efficiency. Firstly, its computational complexity (FLOPs) is 151.95 G, which is lighter than many mainstream models, attributable to its efficient backbone design. Secondly, the model exhibits high training efficiency, requiring only 2.5 hours to complete training, thereby significantly accelerating the research iteration process. Finally, during the inference phase, the model achieves a processing latency of 158.24 ms (approx. 6.32 FPS) per image. While this speed may not be suitable for real-time video streams, it is fully sufficient for clinical offline analysis and batch data processing, demonstrating the model's practical effectiveness and feasibility.

D. MODEL STATISTICAL VALIDATION

To rigorously assess the performance and reliability of our proposed model, we conducted comprehensive statistical validation on the IDRID dataset. The evaluation focused on four primary lesion types—MA, HE, EX, and SE—across four independent experimental runs. The performance was quantified using three key metrics: AUPR, DICE, and IOU. The average performance and standard deviations are presented in Table 8. The experimental results indicate a consistent performance hierarchy across the different lesion types. The model achieves superior performance when segmenting larger and more distinct lesions; exudates (EX) obtain the highest scores on all metrics (AUPR = 69.13%, DICE = 83.09%, IoU = 67.66%). Hard exudates (SE) follow closely, indicating the model's strong capability in handling high-contrast targets. Conversely, the model encounters greater difficulty with smaller and less distinct lesions. Hemorrhages (HE) show moderate performance, while MA

TABLE 6. Impact of different input sources and backbone networks on segmentation results (Bold indicates the best performance).

Methods	Green Encoder	Input	mAUPR	mDice	mIOU
Baseline	UNet	Whole	60.06	57.14	39.43
Model1	ResNet50	Whole, Green	61.56	62.28	47.30
Model2	VGG16	Whole, Green	60.90	57.63	42.80
Model3	DeepLabV3+	Whole, Green	63.79	59.54	44.23
Our Method	EfficientNetV2-S	Whole, Green	70.80	68.61	53.24

TABLE 7. Computational efficiency on the IDRID dataset.

Metric	Value
FLOPs	151.95 G
Training Time	2.5 h
Test Time	158.24 ms
FPS (on CUDA)	6.32

TABLE 8. Average performance of four lesions over four experiments on the IDRID dataset.

Lesion	AUPR	DICE	IOU
MA	55.46 ± 0.015	53.84 ± 0.0101	37.43 ± 0.0248
HE	62.52 ± 0.018	63.54 ± 0.0085	45.23 ± 0.0232
EX	69.13 ± 0.092	83.09 ± 0.0205	67.66 ± 0.0179
SE	67.76 ± 0.073	76.35 ± 0.0201	57.90 ± 0.0099

yield the lowest scores (AUPR = 55.46%, DICE = 53.84%, IoU = 37.43%), which is consistent with the known challenge of detecting minute vascular abnormalities in retinal images. A critical aspect of our validation is the analysis of result stability, as indicated by the standard deviations. Except for the AUPR of EX and SE, all standard deviations are remarkably low (mostly below 0.025), demonstrating high reproducibility and stability of the model's performance across multiple runs. The slightly higher variance observed for the AUPR of EX (± 0.092) and SE (± 0.073) may be attributed to the inherent variability in the presentation and distribution of these lesions within the dataset. In summary, the statistical validation confirms that our model is both effective and reliable, particularly in segmenting lesions with well-defined characteristics, thus laying a solid foundation for clinical application and future research.

VII. CONCLUSION

This paper addresses the challenges in diabetic retinal lesion segmentation, including low contrast, small inter-class differences, and difficulties in segmenting small lesions, by proposing a dual-encoder lesion segmentation network (TEDR-YNet) aimed at enhancing the model's segmentation capability through extracting representations of lesion regions from different input sources. The model extracts multi-representational features through its dual-encoder structure, significantly enhancing the model's expressive capability; introduces a lesion-aware enhancement module in the bottleneck layer to improve multi-scale lesion perception and boundary segmentation capabilities; innovatively embeds a gated convolution mechanism in

skip connections to strengthen focus on lesion regions while suppressing responses in non-lesion areas; further proposes an edge enhancement module that optimizes edge semantic information extraction by simulating the visual system's sensitivity to local contrast; simultaneously designs a hybrid loss function to alleviate class imbalance problems. Experiments on three public datasets (IDRID, DDR, and TJDR) show that TEDR-YNet significantly outperforms baseline models across multiple evaluation metrics, demonstrating excellent segmentation accuracy and robustness, providing reliable technical support for clinical auxiliary diagnosis of diabetic retinopathy. Although TEDR-YNet has achieved good performance on multiple datasets, this study still has certain limitations. The model's recognition stability for subtle lesions such as microaneurysms and small exudates in extremely low-contrast regions still has room for improvement. Furthermore, the current method has not explicitly incorporated clinical semantic priors such as retinal vascular structure and anatomical partitions, which may somewhat limit the model's segmentation consistency and interpretability for complex cases. Future research will attempt to integrate retinal structural constraints and pathological feature correlation modeling to enhance the clinical rationality and reliability of model outputs. Meanwhile, we plan to explore multimodal fusion strategies, combining complementary imaging information such as OCT to construct a more comprehensive lesion representation system.

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